

Software Developer

Candidate ID: 22235

Rockford, IL

- More than 12 years of experience in Real-Time Embedded Software Design, Development, Verification and Validation, Certification activities of airborne systems as per DO-178B/C.
- Experience in every phase of the V-model Software Development Life Cycle (SDLC).
- Experience in all levels of Aerospace Safety Standards DO-178B/C.
- Experience in writing the High Level and Low-Level Software Requirements from Customer Spec and ICDs.
- Experience in Software Development and Verification and Validation using Embedded C and Ada
- Experience in Requirements/Design/Code Reviews & Data/Control Coupling Analysis, Structural Coverage Analysis, Memory Analysis, and Timing Analysis.
- Experience in various IDE's like Code Composer Studio and Green Hills Multi 4.2.
- Experience in communication protocols like ARINC 629, ARINC 429 and eCAN.
- Experience in the usage of various configuration management tools like Clear Case, Tortoise Subversion, DOORS, PVCS Dimensions, Synergy CM and Change Synergy.
- Hands-on experience on MATLAB Simulink and State flow using state machines and flow charts.
- Have overall 4 years onsite experience at various Customer locations in the US. Authorized to work in the US for any employer

WORK EXPERIENCE

Software Developer

HCL | December 2015 to Present

Project: Electric Power Generating System

- High Level requirement development through analysis of the system requirements, system architecture and customer specifications.
- Requirements analysis and create software requirements.
- Development of airborne software code as per DO178B/C standards.
- Development of Matlab Simulink models corresponding to system requirements.
- Integration of software modules, integration testing and Software debugging.
- Reviews of requirements/design/code/models.
- Data/Control coupling analysis, Structural coverage analysis, Memory analysis and Timing Analysis.
- Development of uHSIT Test Procedure and Test Case Scenarios.
- Development of verification activities for the change reports.
- Software verification using Emulator, SITE, uHSIT Test methods

Project Lead

HCL | May 2015 to November 2015

Project: Next Generation Satellite System

- Requirement based Functional testing for Next generation satellite system software.
- Technical reviews of project deliverables including requirements, design, code and verification artifacts.
- Test cases, test procedures and test scripts development as per DO178B/C guideline.
- Integration of software modules, integration testing and software debugging.
- Verification and validation activities for Level-A software per DO178B/C guideline

Project Lead

HCL | December 2014 to April 2015

Project: Integrated Surveillance System

- Requirement based functional testing for Integrated Surveillance System software modules
- Developed test cases, test procedures and test scripts.
- Performed integration of software modules, integration testing and software debugging.
- Validation of software traceability between system requirement, software requirement and design requirements.
- Structural coverage analysis, data and control coupling analysis.
- Reviews of requirements, test artifacts, code, structural coverage, data and control coupling analysis.
- Performed run for score analysis on the Black label build

Project Lead

HCL | December 2010 to December 2014

Project: WEATHER RADAR-2100

- High Level requirement development through analysis of the system requirements and system architecture.
- Analysis of software high level requirements, low level requirements and participated in design activities.
- Development of test cases, test procedures and test scripts as per DO178B/C standards using C and assembly coding.
- Integration of software modules, integration testing and software debugging.
- Data Coupling and control coupling analysis and structural coverage analysis as per DO178B/C guideline.
- Formal technical reviews of project deliverable including requirements, design, code and verification artifacts

Lead Engineer

HCL | March 2010 to November 2010

Project: ARINC 661 Graphics Server

- Requirement based functional testing for ARINC 661 Graphics Server software modules.
- Development of test cases, test procedures and test scripts as per DO178B/C standards using C++.
- Integration of software modules and software debugging.
- Validation of software traceability between software requirement and design requirements
- Reviews of requirements, test artifacts, code, DCCC and Structural coverage analysis.
- Performed run for score analysis

Software Engineer

HCL | May 2008 to February 2010

- High Level requirement development through analysis of the system requirements and system architecture.
- Analysis of software high level requirements, low level requirements and participated in design activities.
- Development of test cases, test procedures and test scripts as per DO178B/C standards using C and Ada.
- Integration of software modules, integration testing and software debugging.
- Data coupling and control coupling analysis and structural coverage analysis per DO178B/C guideline.
- Performed timing, stack and memory analysis per system requirements and product specifications.
- Validation of software traceability between system requirement, software requirement and design requirements.
- Formal technical reviews of project deliverable including requirements, design, code and verification artifacts.
- Performed run for score analysis

Software Engineer

Indage software services and Pvt limited | September 2007 to May 2008

- High Level requirement development through analysis of the system requirements and system architecture.
- Development of test cases, test procedures and test scripts as per DO178B/C standards.
- Integration of software modules, integration testing and software debugging.
- Validation of software traceability between system requirement, software requirement and design requirements.
- Formal technical reviews of project deliverable including requirements, design, code and verification artifacts

EDUCATION

Bachelor's in Electronic Communications Engineering Belgaum, Karnataka

SKILLS

C and Ada (10+ years)

ADDITIONAL INFORMATION

Skills: Programming Languages: C, Ada and Assembly Language Hardware Systems: MPC5554, General Processing Module (GPM), Freescale Microcontrollers, Intel 80C196KC, TI TMS320C44 processor, TITMS320VC33, ADSP 21065L processor and ARM Cortex A8 Communication Protocols: ARINC 629, ARINC 429 and eCAN Standards: RTCA DO-178B and DO-178C Development Tools: Cygwin gcc compiler, uHSIT, Code Composer, Visual Studios, CISS-Interface Tool, CISS-Data Recorder Tool, WxI-708, DataTrack2100, EM420, FlightTestRecorder, PlaybackApp and Green Hills Multi IDE Configuration Tools: Telelogic DOORS, Tortoise Sub-version, ClearCase, PVCS. IBM Rational Synergy Testing Tools: Matlab, Simulink, Ashling Emulator, Pathfinder, Eclipse IDE, Ethernet Manager, Testlink, MI-TERM, Rational Test Real Time (RTRT), VISTA, Understand ADA/ C, Code Composer, Logic Analyzer, ADS-B Sim, Data Recorder, ADSB Sim Playback, LabVIEW simulator, Simgen GSS6700, GNSS Vision and Rule Checker.